modle1

Report
27 May 2022

Survey Data

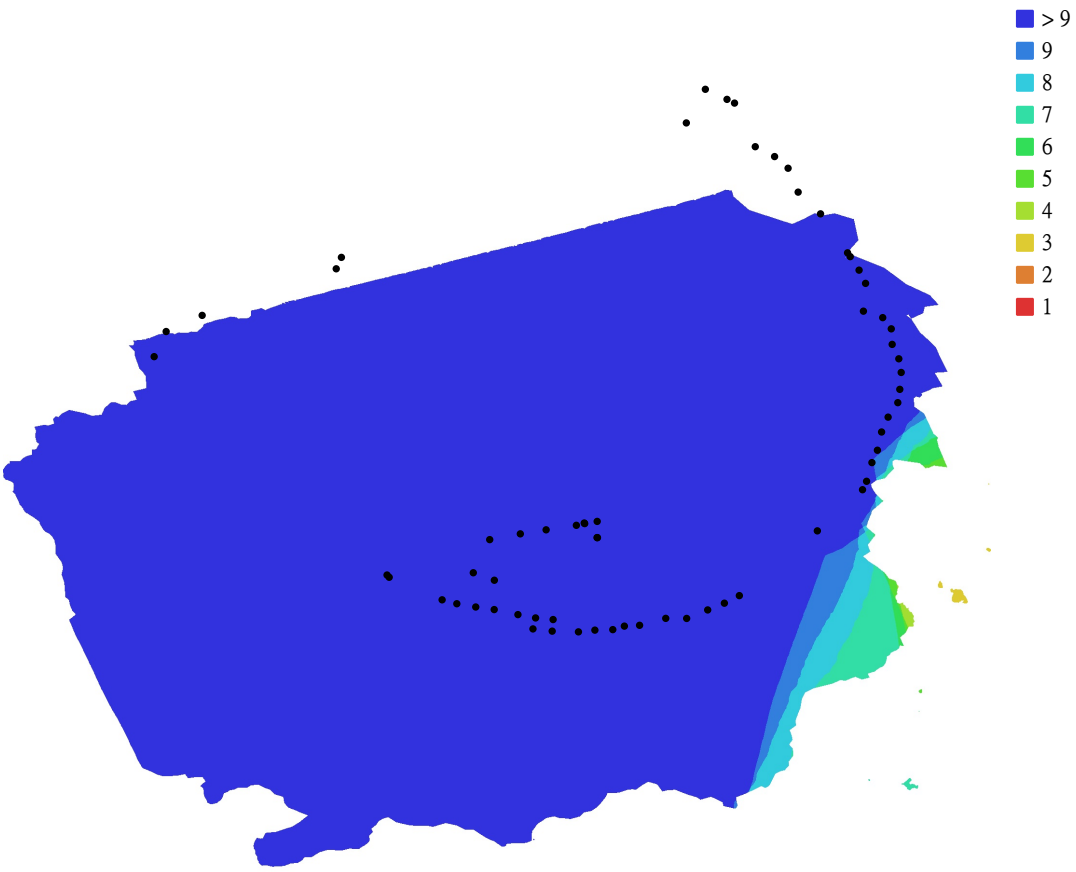

Fig. 1. Camera locations and image overlap.

Number of images:	65	Camera stations:	65
		Tie points:	15,388
		Projections:	120,088
		Reprojection error:	2.41 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
iPhone 11, iPhone 11 back dual wide…	4032 x 3024	4.25 mm	1.4 x 1.4 μm	No

Table 1. Cameras.

Camera Calibration

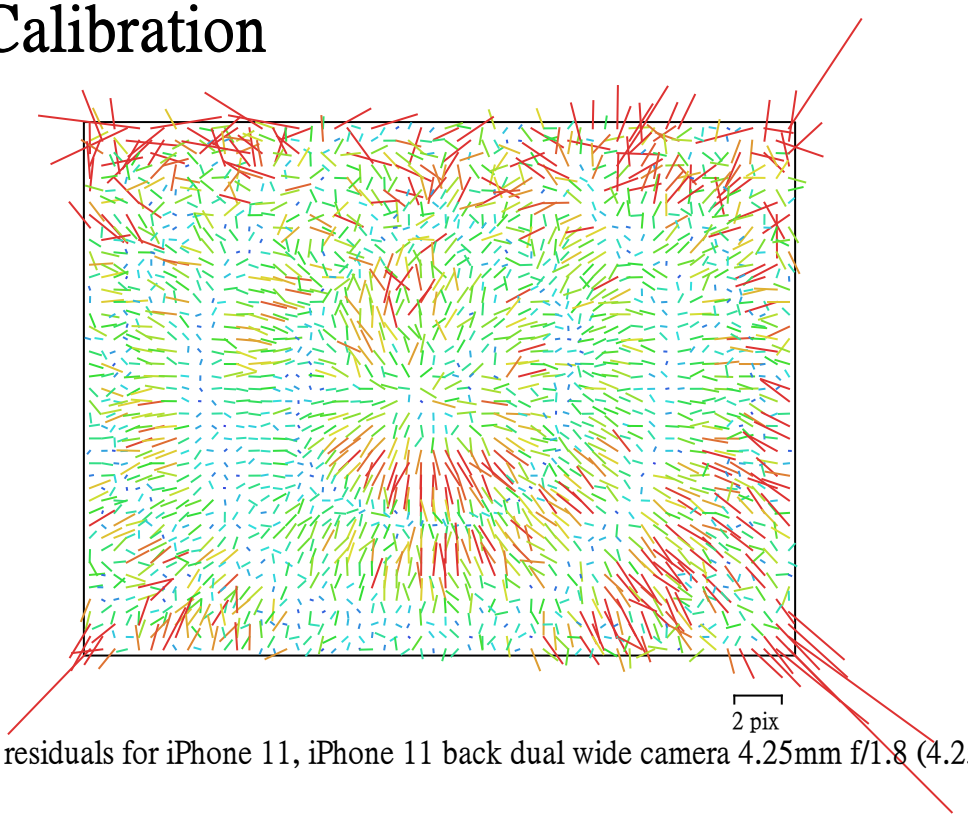

Fig. 2. Image residuals for iPhone 11, iPhone 11 back dual wide camera 4.25mm f/1.8 (4.25mm).

iPhone 11, iPhone 11 back dual wide camera 4.25mm f/1.8 (4.25mm)

65 images

Type	Resolution	Focal Length	Pixel Size
Frame	**4032 x 3024**	**4.25 mm**	**1.4 x 1.4 μm**

	Value	Error	F	Cx	Cy	K1	K2	K3	P1	P2
F	**3072.18**	0.45	1.00	-0.04	-0.74	-0.31	0.26	-0.21	0.01	-0.22
Cx	**-10.7462**	0.52		1.00	0.01	0.08	-0.06	0.05	0.98	0.05
Cy	**-5.95366**	0.61			1.00	-0.02	0.01	-0.02	-0.02	0.57
K1	**0.0742909**	0.00058				1.00	-0.96	0.89	0.07	-0.06
K2	**-0.210975**	0.0019					1.00	-0.98	-0.06	0.01
K3	**0.198505**	0.0021						1.00	0.04	-0.01
P1	**0.00157055**	6.1e-05							1.00	0.02
P2	**-0.00152181**	5.1e-05								1.00

Table 2. Calibration coefficients and correlation matrix.

Digital Elevation Model

Fig. 3. Reconstructed digital elevation model.

Processing Parameters

General	
Cameras	65
Aligned cameras	65
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Point Cloud	
Points	15,388 of 30,907
RMS reprojection error	0.844892 (2.40931 pix)
Max reprojection error	4.97912 (81.1351 pix)
Mean key point size	2.87966 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	9.20303
Alignment parameters	
Accuracy	High
Generic preselection	No
Reference preselection	No
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	1 hours 19 minutes
Matching memory usage	1.04 GB
Alignment time	2 minutes 14 seconds
Alignment memory usage	45.38 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Optimization time	7 seconds
Date created	2022:05:26 14:26:35
Software version	1.8.1.13915
File size	6.98 MB
Depth Maps	
Count	65
Depth maps generation parameters	
Quality	High
Filtering mode	Moderate
Max neighbors	16
Processing time	1 hours 26 minutes
Memory usage	2.79 GB
Date created	2022:05:26 17:19:25
Software version	1.8.1.13915
File size	244.19 MB
Dense Point Cloud	
Points	7,415,275
Point colors	3 bands, uint8
Depth maps generation parameters	
Quality	High
Filtering mode	Moderate

Max neighbors	16
Processing time	1 hours 26 minutes
Memory usage	2.79 GB
Dense cloud generation parameters	
Processing time	1 hours 24 minutes
Memory usage	3.02 GB
Date created	2022:05:26 18:44:00
Software version	1.8.1.13915
File size	198.14 MB
Model	
Faces	351,466
Vertices	176,586
Vertex colors	3 bands, uint8
Texture	1,024 x 1,024 x 4, 4 bands, uint8
Depth maps generation parameters	
Quality	High
Filtering mode	Moderate
Max neighbors	16
Processing time	1 hours 26 minutes
Memory usage	2.79 GB
Reconstruction parameters	
Surface type	Arbitrary
Source data	Dense cloud
Interpolation	Enabled
Strict volumetric masks	No
Processing time	14 minutes 54 seconds
Memory usage	2.37 GB
Texturing parameters	
Mapping mode	Generic
Blending mode	Mosaic
Texture size	1,024
Enable hole filling	Yes
Enable ghosting filter	No
UV mapping time	1 minutes 46 seconds
UV mapping memory usage	182.02 MB
Blending time	1 minutes 28 seconds
Blending memory usage	665.84 MB
Date created	2022:05:27 01:29:06
Software version	1.8.1.13915
File size	20.41 MB
System	
Software name	Agisoft Metashape Professional
Software version	1.8.1 build 13915
OS	Windows 64 bit
RAM	7.88 GB
CPU	Intel(R) Core(TM) i7-5500U CPU @ 2.40GHz
GPU(s)	Intel(R) HD Graphics 5500